

**Under the course of Climate Change,
Sustainability and Development**

A Report On

**CLIMATE
INEQUALITY
&
EMISSIONS
ANALYSIS FOR THE
UNITED STATES**

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1. Country Profile and Data Compilation

1.1 Country Overview

The United States of America is among the most important examples for analysing global CO₂ emissions. It is the world's largest historical producer of carbon dioxide and the second-largest current producer, behind China (Environmental Protection Agency [EPA], 2025). The United States produces 12.91% of the worldwide CO₂ emissions, which is disproportionate and lies at the core of the analytical structure of this report (Global Carbon Project, 2025a).

The United States is a high-income nation economically, with a GDP per capita of \$81,032 in 2023 (World Bank, 2024a). It has experienced a structural change in the last 20 years, where its economy has shifted from a manufacturing-based model to a service and technology-intensive model. The US energy sector has traditionally relied on fossil fuels, especially coal, natural gas, and petroleum, but the decarbonisation of the electric power sector has been much more rapid than the transport one (U.S. Energy Information Administration [EIA], 2021). The United States has contributed about 25% of all CO₂ emissions since industrialisation. This directly informs climate responsibility questions and the principle of common but differentiated responsibilities (CBDR) in the United Nations Framework Convention on Climate Change (UNFCCC) (Chancel, 2022; Rajamani, 2000).

1.2 Dataset and Sources

The dataset timeline used for this report is 2004-2023, which is 20 years of confirmed emissions and development data. This timeframe was chosen because it includes the post-Kyoto Protocol period, the energy transition due to the shale gas boom, the Global Financial Crisis (GFC) of 2008-2009 and the 2020 COVID-19 anomaly and recovery.

The data on the CO₂ emissions, total and per capita, was taken out of the Global Carbon Budget v15 (2025), which is available on Our World in Data (OWID). The figures of the population and GDP per capita were obtained through the World Development Indicators provided by the World Bank. The data on the sector-wise emissions were obtained through the Inventory of US Greenhouse Gas Emissions and Sinks: 1990-2023 (EPA 430-R-25-003). The same OWID dataset was also used to provide comparative per capita values of India, China, and the World Average, providing the same methodology to all comparison series.

The dataset includes the following variables:

Variable	Unit	Source
Total CO ₂ Emissions	Million Tonnes (Mt)	Global Carbon Project (2025) via Our World in Data
Population	Persons	World Bank: SP.POP.TOTL (2024)
Per Capita CO ₂ : US	Tonnes/person	Global Carbon Project (2025) via Our World in Data
Per Capita CO ₂ : India, China, World	Tonnes/person	Global Carbon Project (2025) via Our World in Data
GDP per Capita	Current USD	World Bank: NY.GDP.PCAP.CD (2024)
US Share of Global CO ₂	%	Calculated: US Total ÷ World Total × 100
Sector-wise Emissions	MMT CO ₂ eq.	EPA Inventory of US GHG Emissions: 1990–2023 (2025)

Table 1: Dataset variables, units, and sources.

2. Emissions Trend Analysis

2.1 Total CO₂ Emissions Trend (2004–2023)

Figure 1 shows the trajectory of the United States total CO₂ emissions from fossil fuels and industry across the 20-year study period. The overall trend moves downward from 6,111.55 million tonnes (Mt) in 2004 to 4,918.41 Mt in 2023 with a reduction of 19.5% over two decades.

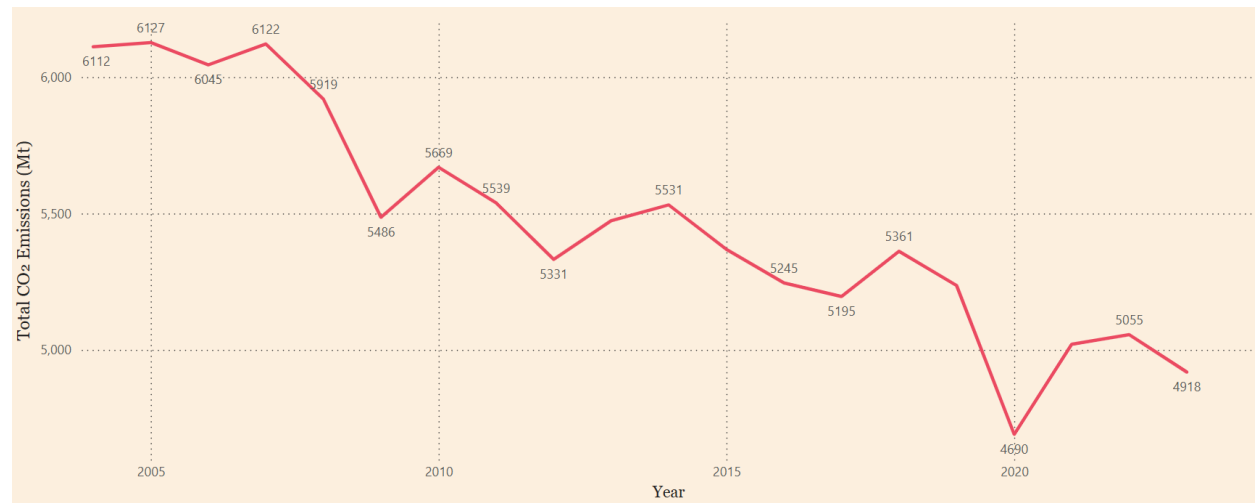


Figure 1: US Total CO₂ Emissions (2004–2023). Source: Global Carbon Project (2025) via Our World in Data.

The trend reveals three different phases. A relatively stable plateau is shown by the first phase (2004–2007), during which time the emissions fluctuated between 6,045 and 6,122 Mt on average per year. These years also saw uninterrupted GDP growth. In the era of President George Bush, the economy grew at a rate of 2–3% per year, which made it possible to sustain the energy demand of all the sectors.

The onset of the 2008 Global Financial Crisis marked the start of the second phase. Emissions dropped dramatically from 5,919 Mt in 2008 to 5,486 Mt in 2009, i.e., by 433 Mt in one year. It is one of the largest declines in emissions in US history, apart from the one during the pandemic year that mainly affected industrial output, travel and energy demand. The partial recovery thereafter (2010–2014) coincides with the resumption of the economy's growth, though the emissions have not reached the pre-crisis levels.

The last phase corresponds to the period when coal plant closures were rapidly increasing, and the number of cheap natural gases obtained through hydraulic fracturing (fracking) was also increasing, leading to the substitution of coal by natural gas in the electricity generation mix (2015–2019). The EIA (2021) states that the coal-to-gas switch accounted for about 65% of the 819 Mt of US power

sector CO₂ emission reduction from 2005 to 2019, while the introduction of renewables accounted for the remaining 30%. Chen et al. (2022) point out that it was mainly the switching of one kind of fuel to another which brought about the reduction of energy intensity that led to the decoupling of the US during the period. According to Gathrid et al. (2025), coal plant retirements in the US have been heavily increasing since 2010, with the rates of retirement being four to fourteen times higher than the pre-2010 levels. Emissions fell to 4,690 Mt, which is the lowest level in the whole dataset, and it was almost entirely due to COVID-19 lockdowns.

2.2 Per Capita CO₂ Emissions Trend (2004–2023)

Figure 2 shows the per capita CO₂ emission of the United States. The drop in 20.87 tonnes per capita in 2004 to 14.32 tonnes per capita in 2023 is a 31.4% decrease, which is significantly sharper than the 19.5% decrease in absolute emissions.

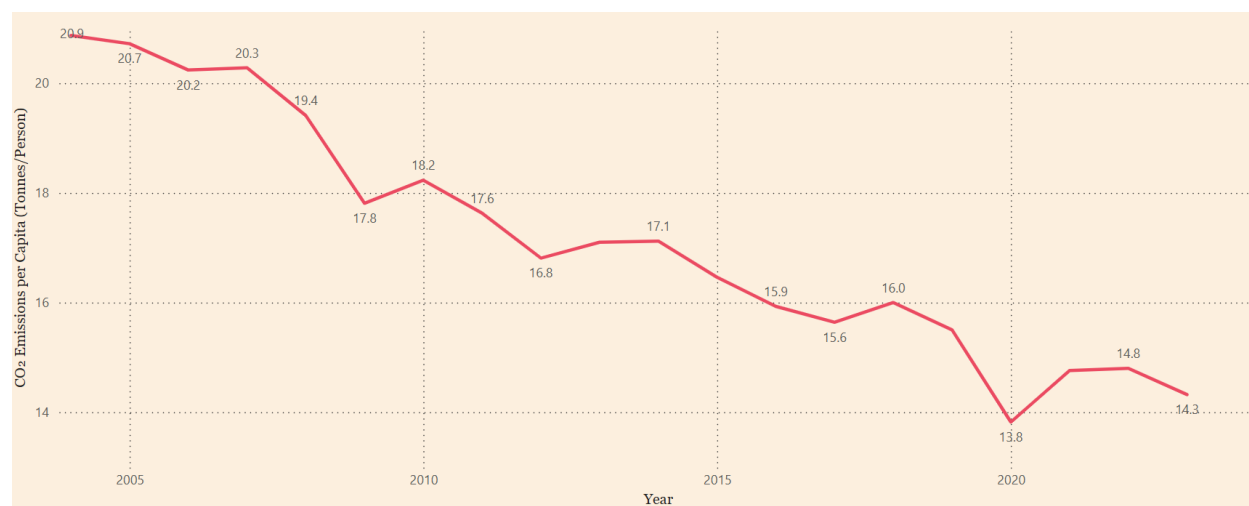


Figure 2: US Per Capita CO₂ Emissions (2004–2023). Source: Global Carbon Project (2025) via Our World in Data.

The graph shows that the 2020 per capita value of 13.82 t/person is the lowest in the dataset and is an anomaly due to the pandemic. Importantly, although a 31.4% per capita decrease is significant, the 14.32 t/person target for 2023 is highly problematic from a climate justice perspective. According to Section 3 of this report, this is 3.04 times the global average (4.71 t/person) and 6.7 times the per capita figure in India (2.13 t/person) during the same year. Progress, then, should not be evaluated against the historical course of the US, but against distributional standards that can be used to determine fair climate responsibility.

2.3 Sector-wise Emissions Breakdown

Figure 3 shows US CO₂ emissions by end-use sector from 2004 to 2023, sourced from the EPA's annual Greenhouse Gas Inventory (EPA, 2025).

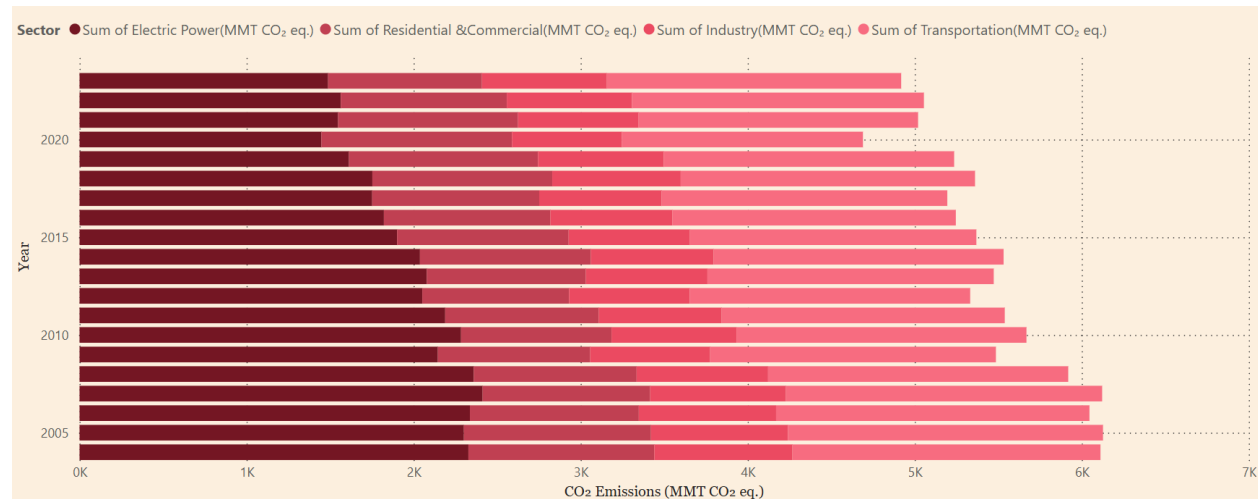


Figure 3: US CO₂ Emissions by Sector (2004–2023). Source: EPA Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2023 (EPA, 2025).

The electric power industry has been the largest contributor to emissions cuts. The emission of electric power decreased by about 840 MMT, or 36%, from 2,328 MMT in 2004 to 1,488 MMT in 2023. This decrease was largely due to the replacement of coal-generated generation by cheaper natural gas produced by the fracking boom and to increased input of wind and solar energy. According to the EIA (2021), in the US power sector, coal-to-gas conversion accounted for about 65% of CO₂ reductions in 2005-2019, while renewable growth accounted for the remaining 30%. According to Gathrid et al. (2025), coal plant retirements in the US have increased rapidly since 2010, and the proportion of electricity generation from coal dropped to less than 20% by 2023.

The transport industry, on the other hand, has been resistant to change. The overall emissions ranged between 1,695 and 1,893 MMT per year throughout the majority of the time. Only the 2020 COVID anomaly (1,444 MMT) caused a significant decrease in almost all forms of transport. By 2023, the transport emissions were back to 1,762 MMT, virtually the same as in 2004, of 1,844 MMT. This indicates that the United States is highly car-dependent, lacks a well-developed transportation network, and has a long history of slower electric vehicle adoption than peer economies. The reality is that the US has been recording the greatest overall emissions cuts, and that transport has changed little since

it became the largest single-sector emitter which indicates that change has been concentrated in the most economically convenient sectors rather than a more transformative energy transition.

The emissions in the industry fell by a moderate amount of 827 MMT (2004) to 748 MMT (2023), in part due to the true efficiency gains but also, in part, due to the offshoring of the manufacturing activity to other economies, like China, which has a direct implication to the distinction between territorial and consumption-based emissions discussed in Section 4.

The Residential and Commercial sector trend is characterised by year-to-year fluctuations that are mainly caused by weather, which are superimposed on a slow underlying trend as building energy efficiency has risen and heating oil is replaced by natural gas in residential applications. The emissions were between about 879 MMT in the light conditions of 2012 and more than 1, 141 MMT in 2020 when the pandemic kept people at home and increased residential energy consumption despite the collapse in commercial activity. The overall stable trend of this industry over 20 years suggests that the increase in building efficiency and fuel switching has been just adequate to cover population and floor area growth, but not enough to cause the type of structural decline experienced in the electric power industry.

3. Carbon Inequality Analysis

3.1 Per Capita Emissions and Country Classification

The carbon inequality study helps us understand the US's emission patterns in a global context by comparing its per capita emissions with those of India, China, and the world average. This study shows that the US is a big polluter by any measure, whether it is total emissions, per capita emissions, share of global emissions adjusted for population, or historical total emissions.

Table 3 and Figure 4 present the cross-country comparison of per capita CO₂ emissions and related development indicators for the year 2023, drawn from the compiled dataset.

Country / Region	Per Capita CO ₂ (t/person) 2023	Total CO ₂ (Mt) 2023	Global Share (%) 2023	GDP/Capita (USD) 2023	Emitter Class
United States	14.32	4,918.41	12.91%	\$81,032	High
China	8.56	11,397.44	29.92%	\$12,597	High
World Average	4.71	38,094.04	100%	—	—
India	2.13	2,830.16	7.43%	\$2,389	Low-Moderate

Table 3: Cross-country emissions and development comparison, 2023. Sources: Global Carbon Project (2025) via Our World in Data; World Bank (2024a, 2024b).

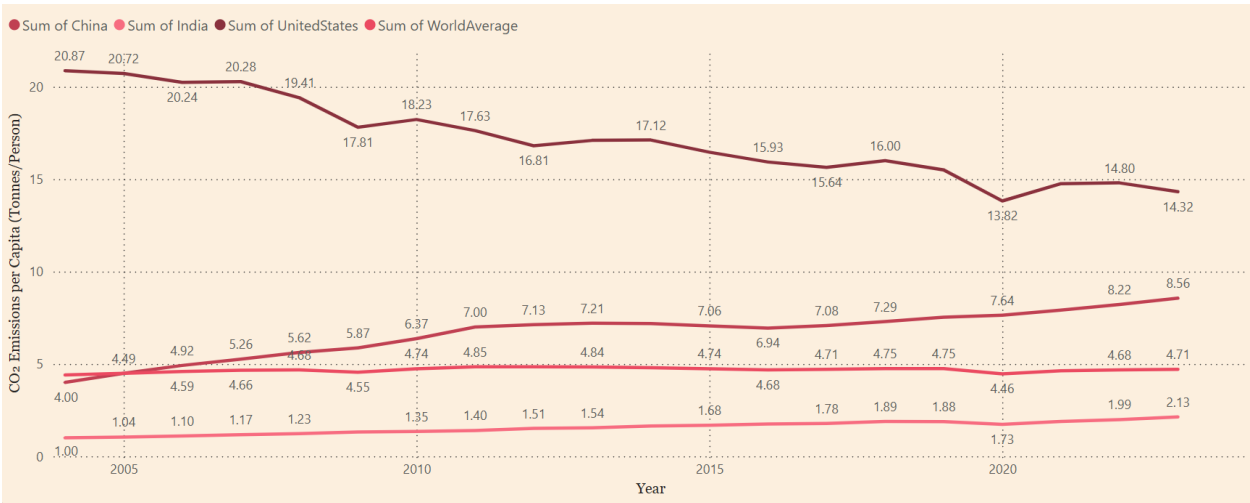


Figure 4: Per Capita CO₂ Emissions: United States, India, China, and World Average (2004–2023). Source: Global Carbon Project (2025) via Our World in Data.

In 2023, US per capita emissions were 14.32 t/person, which was 3.04 times the world average of 4.71 t/person and 6.72 times that of India, which had only 2.13 t/person. Even China, the world's biggest emitter, currently emits 8.56 t/person, which is 40% less than the US per person.

Therefore, the United States qualifies as a high emitter both in absolute and per capita terms. IPCC mentioned that per capita emissions should be reduced to approximately 1.9 t/person worldwide by 2050 if net-zero goals compatible with 1.5°C warming are to be attained (Chancel, 2022). The US, at 14.32 t/person, will have to cut its per capita emissions by about 87% from 2023 levels to meet this target.

According to Chancel (2022), the global bottom 50% of emitters, the group India broadly belongs to on a per capita basis, was responsible for only 16% of global emissions growth between 1990 and 2019, while the global top 1% was responsible for 23%. The average American likely belongs to the top 10% of the world's population by income and accounts for 48% of the world's total greenhouse gas emissions. Ritchie (2023) reports that high- and upper-middle-income countries emit more than 80% of global CO₂ emissions.

3.2 Share of Global Emissions

Figure 5 illustrates the United States' share of global CO₂ emissions from 2004 to 2023. This share decreased from 21.36% in 2004 to 12.91% in 2023, a 40% decrease in proportional share over the period.

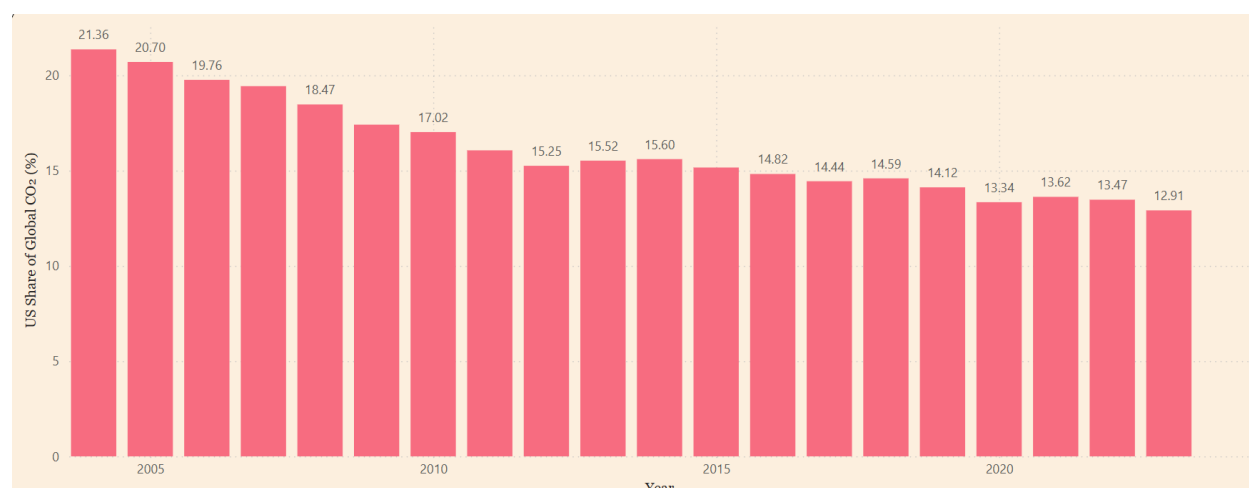


Figure 5: US Share of Global CO₂ Emissions (2004–2023). Source: Global Carbon Project (2025) via Our World in Data.

The reason for the drop in the US share is that the US made real emissions cuts. In contrast, global emissions expanded largely in developing regions, especially China and India, thereby increasing the ratio against which the US share is measured. In 2004, global emissions were 28,610 Mt, 33% more than 2023's 38,094 Mt. To compare, US emissions decreased by 19.5%. The US, which accounts for only 4.2% of the world's population and produces 12.91% of the world's CO₂ emissions, is generating emissions that exceed its population share by a factor of 3.

3.3 Comparison with India and Global Average

The comparison between the United States and India deserves particular analytical attention. India is one of the most populous countries in the world, with a fast-growing economy and is highly vulnerable to climate change. However, its per capita emissions in 2023 were 2.13 t/person, up from 1.00 t/person in 2004 as India moves through its development curve.

Figure 4 shows the per capita trend of the United States, India, China, and the World Average throughout the entire study period. The US line decreases from 20.87 to 14.32 t/person; the India line increases from 1.00 to 2.13 t/person. The ratio between them fell from 20.87x in 2004 to 6.72x in 2023 (a one-third reduction over 20 years). Even with two decades of decreases, the United States is still 6.72 times more carbon-intensive per person than India in 2023. It is one of the core aspects of global carbon inequality that Chancel (2022) describes: the fact that the nations that have contributed the most to atmospheric CO₂ levels still emit more than the global average, despite their pretensions to improvement.

The 20-year global average of 4.71 t/person stability (4.40 t/person in 2004 to 4.71 t/person in 2023), indicates the near-cancellation of US and European declines as compared with increases in China, India, and other developing economies. The US produces 3.04 times the global average despite its emissions cuts. This highlights the historical emissions benefits that rich countries have accumulated and the change that remains to be achieved. This image is further complicated by Hertwich and Peters (2009), who state that the entire emissions footprint of high-consumption economies is not reflected in the territorial emissions, which form the basis of all the figures in this dataset.

4. Visualisation and Interpretation

4.1 Dashboard Overview and KPI Cards

The Power BI dashboard created for this assignment consists of six analysis charts and four summary KPI cards displayed on a single coherent page, allowing observation of the United States' emissions trend across the entire study period. The four KPI cards give the key headline numbers for 2023: per capita CO₂ emissions of 14.32 t/capita, total CO₂ emissions of 4,918 Mt, the US-to-India ratio of 6.70x, and the global share of 12.91%.

The six charts have been arranged in a sequential analytical narrative. There is a trend line of total and per capita emissions (Figures 1 and 2). Figure 3 (sector breakdown) shows the structural drivers. Figure 4 (per capita comparison across four countries) positions the US within the global inequality landscape. The proportional disproportion over time is measured in Figure 5 (global share bar chart). The argument for decoupling is made in Figure 6 (a scatter plot of GDP per capita vs. per capita CO₂). The dashboard is submitted as a .pbix file accompanying this report.

4.2 Scatter Plot: GDP per Capita vs CO₂ per Capita

Figure 6 presents a scatter plot with GDP per capita on the horizontal axis and per capita CO₂ emissions on the vertical axis, with each data point representing one year in the 2004–2023 period.

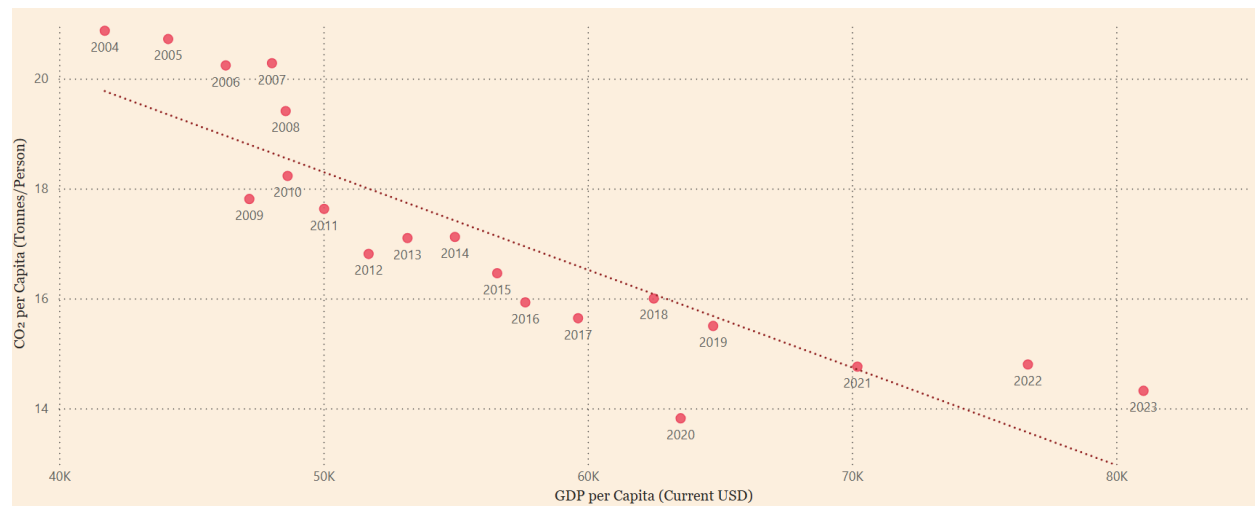


Figure 6: GDP per Capita vs CO₂ per Capita — United States (2004–2023). Source: Global Carbon Project (2025) via Our World in Data; World Bank (2024a).

The scatter plot shows that GDP per capita increased from \$41,724 in 2004 to \$81,032 in 2023 (a 94% increase). At the same time, per capita CO₂ emissions fell from 20.87 to 14.32 t/person (a 31.4% decrease). The data show that as the country became richer, its economy became less carbon-intensive.

Freire-González et al. (2024) examine decoupling dynamics across 164 countries from 1822 to 2018 and find that, although the link between GDP per capita and CO₂ emissions is weakening globally, only 49 countries have achieved absolute decoupling. This is the evidence that puts the US in a small group of countries that have really managed to reduce total emissions by one-third while still growing economically, with almost double its GDP per capita in 20 years.

By definition, "Decoupling" refers to the separation of economic growth from rising CO₂ emissions; a condition in which GDP increases while emissions decline. On the other hand, we cannot take this decoupling at face value; we must critically examine the context, and two main points must be made. First, as the authors Freire-González et al. (2024) point out, decoupling can occur from time to time and should not be expected to happen automatically. For example, the structural factors that made the coal-to-gas switch possible in the US might not necessarily occur on their own in the car sector, where decoupling still needs to happen. Second, in their paper, Hertwich and Peters (2009) have shown that a part of the so-called territorial decoupling in rich countries is due to the shifting of manufacturing emissions to the developing countries rather than efficiency gains. Therefore, some part of the CO₂ decrease visible in the scatter plot might be due to displacement of emissions rather than genuine elimination. Both points do not deny the evidence of decoupling, but are needed for a truthful analytical interpretation of that scatter plot.

5. Climate Justice Reflection

The facts presented in this report make one thing very clear: the United States must take a bold leadership role in global climate mitigation, not as a mere participant, but as the country most responsible both in terms of capacity and historic responsibility.

Capacity-wise, the issue is simple. Besides the US having unparalleled financial and technological resources, as the report notes, the CBDR principle under UNFCCC Article 3 stipulates that developed countries should lead mitigation efforts because of their capabilities (Rajamani, 2000). So, it is impossible to argue that the US cannot afford it financially.

From a responsibility perspective, it is clear that America has the highest obligation. It is responsible for about 25% of all the CO₂ accumulated since the time of industrialisation (Chancel, 2022). In 2023, the per capita emissions gap between the US and India was 6.72 times, which is not just a matter of India's development status. The US has higher per capita emissions, a structural carbon advantage it has already enjoyed. In contrast, India is still trying to attain its share of economic development. This very difference is what the principle of CBDR is supposed to address.

The final point comes from Figure 6. Over the past 20 years, empirical data show that it is possible for economic growth and carbon emissions to decline simultaneously. This evidence of decoupling not only obliges but also guides the provision of funds and technology that help developing nations achieve this new model of transition without having to go down the route of historical emissions and financial capacity requirements (Freire-González et al., 2024). 115 of 164 countries have not decoupled yet, and the US is among the few that have both the means and institutions to bring about change in this regard.

India has 2.13 t/person, versus the US at 14.32 t/person, suggesting that the difference is not one that development will naturally close. Rather, it is a carbon injustice, and to fix it, the United States has to not only step up its pace but also drastically reduce its emissions and invest so that a global transition is possible.

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